Title	Authors	Year	Dataset	Application area	Method / Model	Metrics	Strengths	Limitations	Research gap	Relation to our work
An MRI Scan Brain Tumor Detection Using Deep Learning Techniques (IEEE ICISCoIS)	C. Sivamurugan, E. M. Reddy, T. Manohar, A. Gayatri, D. Revathi, V. Sivakumar	2026	Kaggle Brain MRI dataset; about 1,300 labelled images stated; 4 classes (glioma, meningioma, pituitary, no tumor). Split 80/10/10 (train/val/test).	Multi-class brain tumor classification from MRI with explainability and web deployment	CBAM-Hybrid Net: features from ResNet50V2, DenseNet121, MobileNetV2 and EfficientNet, refined by CBAM attention (channel + spatial), fused (concatenation or voting), Dense 128→64 + dropout, softmax. Preprocessing: 224×224, CLAHE, Gaussian/median filter, augmentation. Grad-CAM heatmaps; Flask web app.	Reported: accuracy 97.8%, precision 0.97, recall 0.98, F1 0.98. Baselines: ResNet50V2 93.6%, DenseNet121 95.1%, MobileNetV2 93.8%, EfficientNetB0 94.7%. Recomputed from the paper's confusion matrix: 380/394 = 96.4%.	Attention plus multi-backbone fusion. Separate validation set. Grad-CAM and a working Flask app. Direct comparison with four single-model baselines.	Test set (394 images) does not fit a ~1,300-image dataset. Reported accuracy differs from the confusion matrix. Fusion method not specified; model named inconsistently; flow diagram mentions binary output and LIME. No Grad-CAM figure, ablation, cross-validation or inference time. Irrelevant literature (KeyBERT, HDBSCAN).	Ensemble cost (time, memory) not measured. No ablation of CBAM or fusion. No patient-level split or external validation.	Reference for attention + fusion design and for a 4-class baseline set. We re-run similar baselines on our own data instead of reusing its numbers; the recomputed 96.4% is the safer reference value.
Multi-Class Brain Tumor Detection Using CNN-Based Medical Imaging Analysis (IEEE ICDT)	L. Singh, A. Banerjee, A. H. Wani, H. Anandaram, A. Nagasri, B. Singh	2025	Online datasets plus "real-time" data; dataset not named; 4 classes (glioma, meningioma, pituitary, healthy). Stratified 70/30 split; test set about 250 per class (our reading of the confusion matrices).	Multi-class brain tumor classification from MRI	Transfer learning with ImageNet-pretrained VGG16, VGG19, ResNet50 and DenseNet169; final layers fine-tuned. 256×256 input, CLAHE, Gaussian/median filtering, augmentation; Adam, cross-entropy, dropout + L2, early stopping.	Accuracy / precision / recall / F1 (%): DenseNet169 97.45 / 97.60 / 97.30 / 97.45 ResNet50 92.34 / 92.50 / 92.20 / 92.34 VGG19 90.23 / 90.40 / 90.10 / 90.23 VGG16 88.65 / 88.80 / 88.50 / 88.65 Recomputed for DenseNet169 from reported counts: about 96.4%.	Four architectures compared in one setup. Full precision/recall/F1 and per-class confusion matrices. Transfer learning, regularisation, early stopping. Stratified split.	Dataset unnamed; counts in the text do not match the reported accuracy. Accuracy and F1 are identical for every model. No separate validation set although early stopping is used (leakage risk). No cross-validation, hyperparameter values or inference time despite a real-time claim. No comparison with prior work.	Patient-level split and external validation. Newer architectures (EfficientNet, ViT). Explainability.	Provides the VGG/ResNet/DenseNet baseline set and a training recipe. Its numbers cannot be used directly; we re-run the same architectures on our dataset.

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Brain Tumor Detection Using Deep Learning-Based CNN Model (IEEE SENNET)	D. S. Keerthan, M. Reddy, C. Athi, I. Singh	2025	BraTS and Kaggle MRI datasets (T1, T2, FLAIR; glioma, meningioma, pituitary, normal). Dataset size and split ratio not stated.	Brain tumor segmentation and classification from MRI	Two-stage pipeline: U-Net tumor segmentation, then feature extraction (GLCM, HOG, LBP) with PCA/LDA reduction, then an EfficientNet-based CNN classifier. Preprocessing: filtering, normalisation, CLAHE, augmentation.	Segmentation mean IoU: training 0.7917, validation 0.3723. Classification accuracy, precision, recall and F1 are printed as "XX%" (placeholders). Confusion-matrix heatmap approx. [[230, 2], [0, 230]] (binary, rounded values).	Segmentation and classification in one pipeline. IoU reported quantitatively. Uses the BraTS benchmark. Predicted vs ground-truth masks shown; overfitting acknowledged.	Final classification metrics missing. Large train/validation IoU gap (0.79 vs 0.37), i.e. weak generalisation. Accuracy/loss curves cover only 3 epochs and fluctuate strongly. Unclear whether handcrafted features or the CNN feed the classifier; class definitions inconsistent (benign/malignant, three types, binary matrix). No baseline numbers; some references unrelated.	Segmentation generalisation across data. Complete classification metrics. Joint end-to-end training of segmentation and classification. Cross-dataset evaluation.	Shows a two-stage segmentation + classification design and its pitfalls. Not usable as a numeric benchmark because the key metrics are missing.
Brain Tumor Detection from MRI (IEEE ICAC2N)	Anshul, A. K. Gupta, D. Maurya, B. Pratihar, A. Khan	2024	Dataset not named (Kaggle-type source mentioned); 80/20 train/test split. Binary: tumor / non-tumor.	Binary brain tumor detection from MRI	Grayscale conversion, noise filtering, skull masking, wavelet and GLCM features, segmentation (pixel / edge / region based), then a custom CNN (two conv + ReLU, max-pooling, further conv, dense layer with sigmoid).	Accuracy 95% (proposed CNN + segmentation). Cited baselines: SVM 85% (Lee & Kim, 2021), DNN 88% (Patel et al., 2022). Precision and specificity appear only in a bar chart.	Clear end-to-end pipeline. Comparison table of segmentation techniques. Data augmentation; simple, reproducible binary task.	Dataset not named; no sample counts. Baseline accuracies come from other studies on different data, so the comparison is not like-for-like. Table I citations do not match the reference list; some references (water, environment) are unrelated. No recall, confusion matrix, cross-validation, ablation or hyperparameters.	Same-dataset comparison with other models. Multi-class classification. Ablation of the segmentation step. Modern architectures and explainability.	Example of a binary CNN baseline with segmentation. Its 95% is not reused as a benchmark; we compare only against numbers we reproduce.

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Brain Tumor Detection with MRI Images (IEEE CISES)	H. Yadav, S. Singh, K. K. Mishra, S. Srivastava, M. S. Naruka, S. P. Yadav	2022	Kaggle MRI data; 1,500 tumor + 1,500 healthy images stated (also described as 3,060 images); 64×64 px; binary (Yes/No). Split ratio not stated.	Binary brain tumor detection from MRI with a simple web interface	Transfer learning with Inception-V3, ResNet-50 and VGG-16. BGR→RGB conversion, 64×64×3 resize, augmentation (ImageDataGenerator). A CNN-SVM hybrid is described but not evaluated.	Recall / precision / accuracy / training time (min) / ROC-AUC / F1: Inception-V3: 0.71 / 0.77 / 0.75 / 93.8 / 0.75 / 0.74 ResNet-50: 0.93 / 0.87 / 0.89 / 151.5 / 0.89 / 0.90 VGG-16: 1.00 / 0.93 / 0.96 / 114.1 / 0.96 / 0.97 Check: F1 values agree with precision/recall (VGG-16 gives 0.964).	Training time reported for each model. Three architectures compared in one setup. Table values are largely self-consistent. Overview of classical and DL methods.	Dataset description contradictory (3,000 vs 3,060 images; BraTS modality text). 64×64 resolution; binary only. ROC-AUC equals accuracy for all models. Training curves (Fig. 12–14) do not reach the reported 0.96; validation accuracy unstable. Split, hyperparameters and hardware not given. CNN-SVM never evaluated.	Higher resolution and multi-class tasks. Evaluation of the proposed hybrid. Timing on identical hardware.	Reporting training time next to accuracy is worth copying. VGG-16 is best here but worst in the 2025 ICDT paper, so rankings depend on the setup and we re-run baselines ourselves.

Reported results at a glance (baselines for later comparison)

Paper (year)	Task	Best model	Reported accuracy	Our check
Sivamurugan et al. (2026)	4-class	CBAM-Hybrid Net	97.8%	96.4% from the confusion matrix (380/394); test-set size inconsistent with dataset size
Singh et al. (2025)	4-class	DenseNet169	97.45%	About 96.4% from reported per-class counts; accuracy and F1 identical
Keerthan et al. (2025)	Segmentation + classification	U-Net + EfficientNet-based CNN	Not reported (“XX%”)	Segmentation IoU 0.79 (train) vs 0.37 (validation)
Anshul et al. (2024)	Binary	Custom CNN + segmentation	95%	Not verifiable; baselines taken from other datasets
Yadav et al. (2022)	Binary, 64×64	VGG-16	96%	Table consistent; training curves stay below the reported value